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personal-portfolio-cyan-eight-50.vercel.app

Summary

Data Engineer with 2 years at Caterpillar who recently shipped a production LLM-powered Anomaly Detection Platform with Cortex Analyst and a three-tier approval workflow. Builds and evaluates GenAI systems outside work: Text-to-SQL with adversarial benchmarking, a hybrid RAG pipeline with RAGAS eval gates, and a quantile MLOps forecasting platform. B.Tech AI/ML (CGPA 8.55).

Work Experience

Caterpillar Inc. Bengaluru, India

Data Engineer Jun 2025 – Present

Promoted from Systems Data Support Analyst within 12 months.

- Architected and shipped the LPAgent Anomaly Detection & Alerting Platform (Streamlit-in-Snowflake), the first production LLM-powered application in PSD at Caterpillar; **Cortex Analyst** generates anomaly rule SQL, cutting rule deployment from a multi-day manual process to same-day self-service.
- Implemented three-tier (L1/L2/L3) rule approval workflow with Snowflake-native role-based access control and real-time role resolution; soft-update audit trail captures all role changes with attribution and reason; reviewer email notifications replatformed from Dataiku to a native Snowflake trigger.
- Led the MineStar MSU SAS-to-Snowflake migration end-to-end: refactored 2,025 lines of legacy SAS into 32 optimised Snowflake objects (69% table reduction) across 5 data domains covering ~5,500 global parts; verified cutover with row-level parity reconciliation.
- Own 25+ production ETL/ELT pipelines (SnapLogic, dbt, Snowflake) with refresh SLAs as tight as 20 minutes, powering Power BI dashboards that drive daily shipment-allocation decisions for ~75 supply-chain stakeholders.
- Designed a Goods Receipt dbt pipeline with SCD Type-2 history at supplier/PO grain, enabling point-in-time audit and invoice dispute resolution previously impossible with snapshot-only reporting; 13 data-contract tests; Dynamic Tables auto-refresh every 30 min.
- Audited a Snowflake Cortex NL-to-SQL prototype, identified critical safety gaps, and open-sourced a hardened rebuild with AST-based guardrails and hallucination detection ; see Text-to-SQL project.

Systems Data Support Analyst Jun 2024 – Jun 2025

- Supported production Snowflake/SnapLogic/Tidal pipelines for warehouse, procurement, and finance stakeholders; owned data-quality triage and ad hoc analytical SQL feeding procurement-planning decisions.
- Initiated a multi-year warehouse parts usage analysis (Python, pandas) to surface seasonal demand patterns; built a Random Forest forecasting prototype with time-based train/test split; findings and methodology carried forward into the Data Engineer role.

Software Engineer Intern Jun 2023 – Sep 2023

- Implemented KS-test + PSI drift detection on 3 production Tidal data streams; tuned alert thresholds against 6 months of historical data, reducing false-positive alerts by ~30% and cutting unnecessary pipeline re-runs.

Skills

AI / GenAI: LLM application development, RAG (Retrieval-Augmented Generation), prompt engineering, hallucination detection, embeddings, semantic search, vector databases (Qdrant), BM25/RRF hybrid search, evaluation/evals, RAGAS, Langfuse, Claude Sonnet API, sentence-transformers (BGE-large)

ML / MLOps: LightGBM (LambdaMART, Quantile), MLflow, Evidently AI, scikit-learn, statistics (KS-test, PSI, quantile regression), time series forecasting, FastAPI

Data Engineering: ETL/ELT, data pipeline orchestration, data modeling, Snowflake (Cortex, Dynamic Tables, Tasks, Streams), dbt (SCD Type-2, snapshots, tests), Apache Iceberg, DuckDB, Dagster, Soda Core, SnapLogic, Tidal

Languages: Python, SQL, C++

Analytics & BI: Pandas, Power BI

Infra & Dev: Docker, GitHub Actions CI/CD, Git, PostgreSQL 15

Projects

Text-to-SQL with Guardrails & Hallucination Detection | [GitHub](#)

FastAPI · PostgreSQL 15 · Claude Sonnet · sqlparse · sentence-transformers · Docker

- NL-to-SQL over a supply chain PostgreSQL database; sqlparse AST guardrails (token-type checking, not regex) hard-block DDL/DML, injection, and catalog access; all execution runs inside PostgreSQL SET TRANSACTION READ ONLY as a database-enforced fallback. **100% guardrail block rate on 10 seeded injection tests; zero unsafe queries.**
- 50-question adversarial benchmark (JOINS, CTEs, date arithmetic, ranking, 6 unanswerable queries); back-translation hallucination detection via MiniLM-L6 cosine similarity; multi-query validation surfaces SQL divergence to the user; 6 Architecture Decision Records.

Supply Chain Demand Forecasting MLOps | [GitHub](#) | [Live Demo](#)

LightGBM · MLflow · Evidently AI · FastAPI · Streamlit · Docker · GitHub Actions

- End-to-end forecasting platform for 500 spare-parts SKUs rebuilt from a Caterpillar prototype; QuantileLightGBM (P10/P50/P90) via walk-forward CV achieves **34.9% WAPE and 82.6% interval coverage** (46% over seasonal-naive best baseline); versioned Parquet feature store with 33 hand-crafted features (feature engineering: lag, rolling stats, seasonality, stockout-adjusted).
- MLflow model registry with **promotion gate** (candidate must beat production WAPE by >2%); Evidently AI separates data drift and concept drift with two automated responses; GitHub Actions CI/CD pipeline triggers automated retraining on drift alerts; FastAPI model serving returns P10/P50/P90 with model-version headers.

Hybrid RAG System with Evaluation Suite | [GitHub](#) | [Live Demo](#)

BGE-large · BM25/RRF · Qdrant · RAGAS · Claude Sonnet · Langfuse · GitHub Actions · Docker · Render

- Production RAG pipeline over Snowflake and dbt documentation; dense retrieval (BGE-large, Qdrant) fused with BM25 via RRF (k=60); BGE cross-encoder reranker; Claude Haiku query rewriter; answers grounded with cited source chunks; live on Render.
- 50-question adversarial benchmark across 4 tiers; best configuration achieves **0.86 RAGAS faithfulness** on conceptual queries vs. 0.31 baseline; 3-config ablation quantifies reranker vs. hybrid retrieval contribution; CI eval gate fails build on faithfulness regression; all retrievals traced in Langfuse.

GitHub Archive Lakehouse | [GitHub](#)

Apache Iceberg · DuckDB · Dagster · Soda Core · MinIO · Streamlit · Docker

- Bronze/Silver/Gold Iceberg lakehouse ingesting **7.4M+ GitHub Archive events**; manifest-tracked crash-resumable pipeline; Silver deduplicates by event ID, parses 15 event-type payloads, routes ~600 malformed rows to quarantine with error reason.
- **36 Soda Core data quality checks** gate each Dagster run; GitHub Actions CI/CD pipeline fails on >3 check failures; Software-Defined Assets provide full lineage; Iceberg time-travel demonstrated on a 3.8M-row table; 5 documented failure scenarios with exact recovery commands.

Achievements

- **LeetCode Knight**: max rating 1923; global rank 283/29,865 in Weekly Contest 366 (top 1%) | amitabhchoudhury19999
- **Codeforces**: max rating 1800 (Expert) | amitabh1208
- **CodeChef**: max rating 1711 (4★) | amitabh1208
- **Centurion Hackathon (National Level)**: Team Lead; 13th / 500+ participants.

Education

M.S. Ramaiah University of Applied Sciences, Bengaluru. B.Tech in Artificial Intelligence and Machine Learning, CGPA 8.55/10 2020 – Apr 2024